Chemistry - Graduate Level Assessment 17 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Substances whose Lewis structures must be drawn with an unpaired electron are called
   (a) hydrogen bonds
   (b) free radicals
   (c) allotropes
   (d) isotopes
   (e) nonpolar molecules

2. Consider the Lewis structures for the following molecules: CO 2, CO 32-, NO 2-, and NO 3-. Which molecule would have the shortest bonds?
   (a) O_2
   (b) NO 3
   (c) NO 2-
   (d) CO 32-
   (e) NO 2

3. What is the NMR frequency of 31 P in a 20.0 T magnetic field?
   (a) 310.2 MHz
   (b) 500.5 MHz
   (c) 275.0 MHz
   (d) 400.0 MHz
   (e) 120.2 MHz

4. Monatomic ions of the representative elements are often
   (a) very electronegative
   (b) typically metallic
   (c) highly colored
   (d) mostly gases at room temperature
   (e) often very reactive with water

5. Using fundamental trends in electronegativity and bond strength, which of the following should be the strongest acid?
   (a) H_2Se
   (b) H_2CO_3

(c) H_2S
(d) HBr
(e) H_2O

True / False (10 marks)

Answer True or False

6. True or False: The energy of yellow light from a sodium lamp, with a wavelength of 5890 Å, is 1.5 eV and corresponds to 35.2 Kcal/mole.

7. True or False: The swimmer's body temperature increases due to the warm breeze after exiting the cold water.

8. True or False: The combination of MgBr 2 and Na_2CO_3 produces a net ionic equation with significant ionic species remaining.

9. True or False: The angular-momentum quantum number l for a t orbital is 3.

10. True or False: The thermodynamic equilibrium constant K for the reaction H_2 + X_2 2 HX is given by the ratio K = k_a / k_b.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that contribute to the limiting high-temperature molar heat capacity at constant volume (C_V) of a gas-phase diatomic molecule and explain why it approaches 4.5 R.

12. Discuss the relationship between the de Broglie wavelength of a proton and the potential through which it must fall from rest to achieve a wavelength of 1.0×10^{-10} m.

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